

Life Cycle Assessment (LCA) of Printed Circuit Boards (PCBs)

In this individual project, the greenhouse gas (GHG) emissions throughout the life cycle of a printed circuit board (PCB) manufactured from virgin ores were analyzed and compared against the emissions from its production using secondary sources, such as recycled materials. To conduct this assessment, **OpenLCA** software was utilized, along with the open database **BAFU 2026 v1**, and the **EF 3.1** impact assessment method.

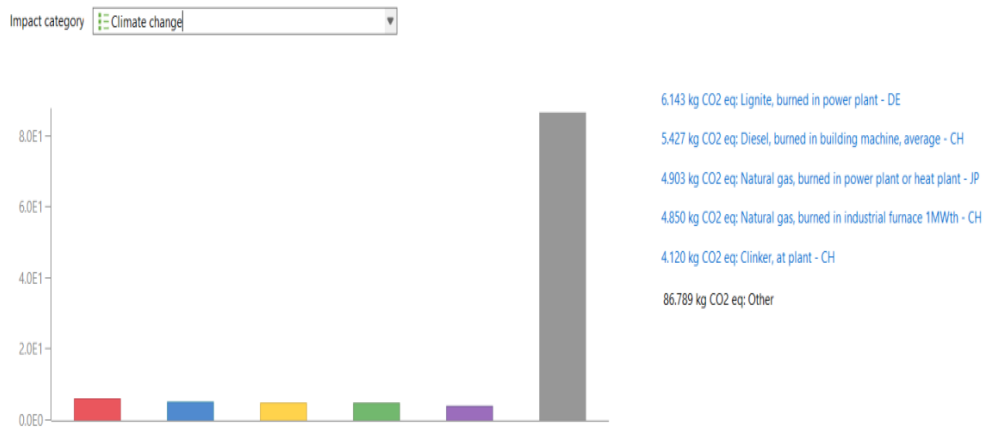


Chart 1: Life cycle emissions diagram of a PCB made from primary materials

According to the chart, the total emissions generated from the manufacturing, use phase, and end-of-life disposal of the PCB amount to **112.232 kg CO2 eq.**

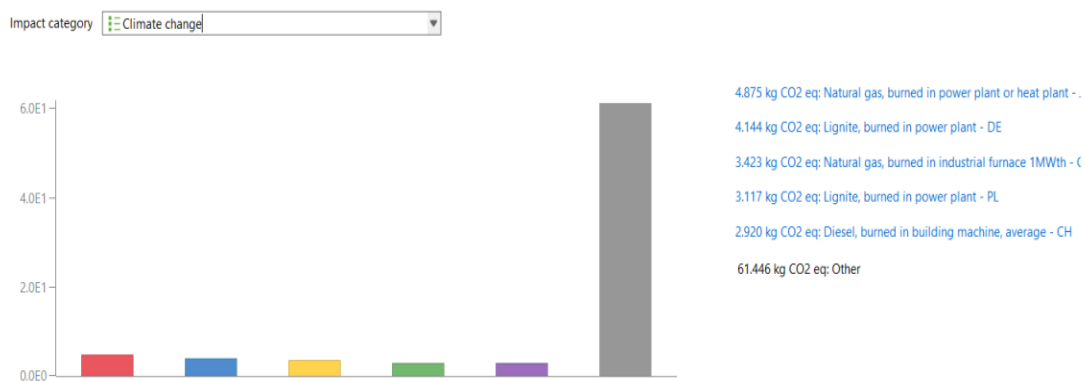


Chart 2: Life cycle emissions diagram of a PCB made from secondary materials

The life cycle emissions of a printed circuit board manufactured from secondary materials total **79.925 kg CO2 eq.**

Conclusion

Implementing eco-design principles in printed circuit boards practically demonstrated the significance of the circular economy, achieving a total reduction of **32.307 kg CO2 eq.** in greenhouse gas emissions. This remarkable outcome highlights the critical need for the industry to transition toward secondary raw materials, eliminating the highly energy-intensive stages of primary mining and ore processing. This environmental saving is equivalent to the carbon sequestration of **1,500 mature trees** in a single year, making it clear that secondary sources are not merely an alternative option, but the most essential strategic tool for radical carbon reduction in global manufacturing.